

SOFTWARE INSTALLATION

Please use the instructions below to install and check your installation.

If you run into issues, you can post a question with information about what you tried and what didn't work, on the [course github discussion page](#) and we will try reply.

Since 29 April 2025, the latest INLA package is built for R 4.5, so if you're able to upgrade your R installation, please do so to avoid unnecessary issues. The package will in many cases also work with older R versions, but compatibility is sometimes difficult.

1 Installing INLA and inlabru

Due to the work involved in building the binaries for the INLA package C software for different architectures, the INLA package is not on CRAN, but it can be installed from its own distribution repository.

1. Check your R version.
2. Install R-INLA, instructions can be found [here](#)
3. Install inlabru (available from CRAN)

```
# Enable universe(s) by inlabru-org
options(repos = c(
  inlabruorg = "https://inlabru-org.r-universe.dev",
  INLA = "https://inla.r-inla-download.org/R/testing",
  CRAN = "https://cloud.r-project.org"
))

# Install some packages
install.packages("inlabru")
```

3. Make sure you have the latest R-INLA, inlabru and R versions installed.
4. Install the following libraries:

```
install.packages(c(
  "dplyr",
  "fmesher",
  "ggplot2",
  "lubridate",
  "magrittr",
  "mapview",
  "patchwork",
  "scico",
  "sf",
  "spatstat",
  "spdep",
  "terra",
  "tidyr",
```

```
"tidyterra",
"tidyverse"
))
```

2 Installation check

Please check your installation using the basic model runs below.

If you run into issues, you can post a question with information about what you tried and what didn't work, on the [course github discussion page](#).

You can check that INLA is correctly installed by running

```
df <- data.frame(y = rnorm(100) + 10)
fit <- INLA::inla(
  y ~ 1,
  data = df
)
summary(fit)
```

Time used:

Pre = 0.399, Running = 0.837, Post = 0.0661, Total = 1.3

Fixed effects:

	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
(Intercept)	9.927	0.109	9.713	9.927	10.14	9.927	0

Model hyperparameters:

	mean	sd	0.025quant	0.5quant	0.975quant	mode
Precision for the Gaussian observations	0.861	0.121	0.64	0.855		
Precision for the Gaussian observations	1.11	0.844				

Marginal log-Likelihood: -162.30

is computed

Posterior summaries for the linear predictor and the fitted values are computed

(Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE)')

If the simple `inla()` call fails with a crash, you may need to install different `inla` binaries for your hardware/software combination, with `INLA::inla.binary.install()`.

When `inla()` works, you can check that `inlabru` is installed correctly by running the same model in `inlabru`:

```
fit <- inlabru::bru(
  y ~ Intercept(1, prec.linear = exp(-7)),
  data = df
)
summary(fit)
```

inlabru version: 2.15.0

INLA version: 26.08.22

Latent components:

```

Intercept: main = linear(1)
Observation models:
  Model tag: <No tag>
  Family: 'gaussian'
  Data class: 'data.frame'
  Response class: 'numeric'
  Predictor: y ~ Intercept
  Additive/Linear/Rowwise: TRUE/TRUE/TRUE
  Used components: effect[Intercept], latent[]
Time used:
  Pre = 0.248, Running = 0.834, Post = 0.0669, Total = 1.15
Fixed effects:
      mean      sd 0.025quant 0.5quant 0.975quant  mode kld
Intercept 9.926 0.109      9.713   9.926      10.14 9.926   0

Model hyperparameters:
                                mean      sd 0.025quant 0.5quant
Precision for the Gaussian observations 0.861 0.121      0.64   0.855
                                0.975quant  mode
Precision for the Gaussian observations      1.11 0.844

Marginal log-Likelihood: -166.76
  is computed
Posterior summaries for the linear predictor and the fitted values are computed
(Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE)')

```